

Introduction to General Relativity: Midterm

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Problem 1. Consider a 2-sphere with coordinates (θ, φ) and a metric

$$ds^2 = d\theta^2 + \sin^2(\theta) d\varphi^2.$$

- (i) Show that lines of constant longitude ($\varphi = \text{constant}$) are geodesics, and that the only line of constant latitude ($\theta = \text{constant}$) that is a geodesic is the equator ($\theta = \pi/2$).
- (ii) Take a vector with components $V_\mu = (1, 0)$ and parallel-transport it once around a circle of constant latitude. What are the components of the resulting vector, as a function of θ ?
- (iii) Consider the following metric for one space-like and one time-like dimension:

$$ds^2 = t^{-2} (-dt^2 + dx^2).$$

Find all the Christoffel symbols, and then calculate the geodesics of this spacetime. *Hint: for this second part, it is easiest to start with the definition of a geodesic as a maximum of proper time and find a relation for x and t by calculus of variance.*

Solution to (i). The geodesic equations are given by:

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0 \quad (1)$$

For the given metric, we can compute the non-zero Christoffel symbols:

$$\begin{aligned} \Gamma_{\varphi\varphi}^\theta &= -\sin(\theta) \cos(\theta) \\ \Gamma_{\theta\varphi}^\varphi &= \cot(\theta) = \Gamma_{\varphi\theta}^\varphi. \end{aligned}$$

Now, let's consider lines of constant longitude, i.e., $\varphi = \text{constant}$. In this case, $\dot{\varphi} = 0$, and the geodesic equations reduce to:

$$\begin{aligned} \ddot{\theta} + \Gamma_{\theta\theta}^\theta \dot{\theta}^2 &= 0 \\ \ddot{\varphi} + \Gamma_{\theta\varphi}^\varphi \dot{\theta} \dot{\varphi} &= 0. \end{aligned}$$

Since $\dot{\varphi} = 0$, the second equation is automatically satisfied. The first equation simplifies to $\ddot{\theta} = 0$, which means that θ changes linearly with the affine parameter, confirming that lines of constant longitude are geodesics.

Next, let's consider lines of constant latitude, i.e., $\theta = \text{constant}$. In this case, $\dot{\theta} = 0$, and the geodesic equations reduce to:

$$\begin{aligned} \ddot{\varphi} + \Gamma_{\varphi\varphi}^\theta \dot{\varphi}^2 &= 0 \\ \ddot{\varphi} + \Gamma_{\theta\varphi}^\varphi \dot{\theta} \dot{\varphi} &= 0. \end{aligned}$$

Since $\dot{\theta} = 0$, the second equation is automatically satisfied. The first equation simplifies to:

$$\Gamma_{\varphi\varphi}^\theta \dot{\varphi}^2 = 0.$$

This implies that either $\dot{\varphi} = 0$ (which is trivial) or $\Gamma_{\varphi\varphi}^\theta = 0$. The latter condition holds only when $\theta = \pi/2$, which corresponds to the equator. Therefore, the only line of constant latitude that is a geodesic is the equator. $\square$

Solution to (ii). Instead of working with covariant components, it's easier to work with contravariant components for parallel transport. For the inverse metric, we have:

$$g^{\theta\theta} = 1 \quad \text{and} \quad g^{\varphi\varphi} = \frac{1}{\sin^2(\theta)}.$$

Raising the indices using the inverse metric components, we have:

$$V^\theta = g^{\theta\theta} V_\theta = 1 \cdot 1 = 1 \quad \text{and} \quad V^\varphi = g^{\varphi\varphi} V_\varphi = \frac{1}{\sin^2(\theta)} \cdot 0 = 0.$$

Thus, we have the initial contravariant components $V^\mu = (1, 0)$. Using the equations for parallel transport of contravariant components:

$$\frac{dV^\mu}{d\lambda} + \Gamma_{\alpha\beta}^\mu V^\alpha \frac{dx^\beta}{d\lambda} = 0 \quad (2)$$

Parametrizing the curve as $x^\mu = (\theta, \varphi(\lambda))$ with θ constant, we have $\dot{x}^\mu = (0, \dot{\varphi})$. Thus, the equations for parallel transport become:

$$\begin{aligned} \frac{dV^\theta}{d\lambda} + \Gamma_{\varphi\varphi}^\theta V^\varphi \dot{\varphi} &= 0 \\ \frac{dV^\varphi}{d\lambda} + \Gamma_{\theta\varphi}^\varphi V^\theta \dot{\varphi} &= 0. \end{aligned}$$

Substituting the non-zero Christoffel symbols from part (i), we get:

$$\begin{aligned} \frac{dV^\theta}{d\lambda} - \sin(\theta) \cos(\theta) V^\varphi \dot{\varphi} &= 0 \\ \frac{dV^\varphi}{d\lambda} + \cot(\theta) V^\theta \dot{\varphi} &= 0. \end{aligned}$$

Letting $\dot{\varphi} = \omega$ (a constant), we can rewrite the equations as:

$$\frac{dV^\theta}{d\lambda} - \sin(\theta) \cos(\theta) V^\varphi \omega = 0 \quad (3)$$

$$\frac{dV^\varphi}{d\lambda} + \cot(\theta) V^\theta \omega = 0. \quad (4)$$

Solving for V^θ from Equation (3), we get:

$$V^\theta(\lambda) = \sin(\theta) \cos(\theta) \omega \int V^\varphi(\lambda) d\lambda,$$

where we're treating $\sin(\theta)$, $\cos(\theta)$, and ω as constants. Substituting this expression for V^θ into Equation (4), we have:

$$\frac{dV^\varphi}{d\lambda} + \cot(\theta) \left(\sin(\theta) \cos(\theta) \omega \int V^\varphi(\lambda) d\lambda \right) \omega = 0 \quad (5)$$

Differentiating Equation (5) with respect to λ , we obtain a second-order differential equation for V^φ :

$$\frac{d^2 V^\varphi}{d\lambda^2} + \omega^2 \cos^2(\theta) V^\varphi = 0.$$

This is a simple harmonic oscillator equation with general solution:

$$V^\varphi(\lambda) = A \cos(\omega \cos(\theta) \lambda) + B \sin(\omega \cos(\theta) \lambda).$$

We now solve for A and B given the initial conditions. At $\lambda = 0$, we have $V^\varphi(0) = 0$, which implies $A = 0$. To find B , we differentiate V^φ :

$$\frac{dV^\varphi}{d\lambda} = B \omega \cos(\theta) \cos(\omega \cos(\theta) \lambda).$$

By Equation (2), at $\lambda = 0$, we have:

$$\left. \frac{dV^\varphi}{d\lambda} \right|_{\lambda=0} + \cot(\theta) V^\theta(0) \omega = 0.$$

Since $V^\theta(0) = 1$, we have:

$$B\omega \cos(\theta) + \cot(\theta)\omega = 0 \Rightarrow B = -\frac{\cot(\theta)}{\cos(\theta)} = -\csc(\theta).$$

Thus, the solution for V^φ is:

$$V^\varphi(\lambda) = -\csc(\theta) \sin(\omega \cos(\theta)\lambda).$$

Plugging this back into Equation (3), we can solve for V^θ :

$$\begin{aligned} V^\theta(\lambda) &= \sin(\theta) \cos(\theta) \omega \int -\csc(\theta) \sin(\omega \cos(\theta)\lambda) d\lambda \\ &= -\omega \cos(\theta) \left(-\frac{\sec(\theta) \cos(\omega \cos(\theta)\lambda)}{\omega} \right) \\ &= \cos(\omega \cos(\theta)\lambda). \end{aligned}$$

Therefore, the components of the resulting vector after parallel transport around a circle of constant latitude are:

$$V^\mu(\lambda) = (\cos(\omega \cos(\theta)\lambda), -\csc(\theta) \sin(\omega \cos(\theta)\lambda)).$$

To express the result in terms of the angle φ , we note that $\varphi = \omega\lambda$. Thus, the final components are:

$$V^\mu(\varphi) = (\cos(\varphi \cos(\theta)), -\csc(\theta) \sin(\varphi \cos(\theta))). \quad \square$$

Solution to (iii). Given the metric, we have:

$$g_{tt} = -t^{-2}, \quad g_{xx} = t^{-2}.$$

The non-zero Christoffel symbols can be computed as follows:

$$\begin{aligned} \Gamma_{tt}^t &= \frac{1}{2} g^{t\mu} (\partial_t g_{\mu t} + \partial_t g_{t\mu} - \partial_\mu g_{tt}) = \frac{1}{2} g^{tt} (-2t^{-3}) = -\frac{1}{t} \\ \Gamma_{tx}^t &= \frac{1}{2} g^{tt} (\partial_x g_{tt} + \partial_t g_{xt} - \partial_t g_{tx}) = 0 \\ \Gamma_{xx}^t &= \frac{1}{2} g^{tt} (\partial_x g_{tx} + \partial_x g_{xt} - \partial_t g_{xx}) = -\frac{1}{2} g^{tt} (2t^{-3}) = -\frac{1}{t} \\ \Gamma_{tt}^x &= \frac{1}{2} g^{xx} (\partial_t g_{xt} + \partial_t g_{tx} - \partial_x g_{tt}) = 0 \\ \Gamma_{tx}^x &= \frac{1}{2} g^{xx} (\partial_x g_{xt} + \partial_t g_{xx} - \partial_x g_{tx}) = \frac{1}{2} g^{xx} (-2t^{-3}) = -\frac{1}{t} \\ \Gamma_{xx}^x &= \frac{1}{2} g^{xx} (\partial_x g_{xx} + \partial_x g_{xx} - \partial_x g_{xx}) = 0. \end{aligned}$$

Thus, the non-zero Christoffel symbols are:

$$\Gamma_{tt}^t = -\frac{1}{t}, \quad \Gamma_{xx}^t = -\frac{1}{t}, \quad \Gamma_{tx}^x = -\frac{1}{t} = \Gamma_{xt}^x.$$

Using Equation (1), we can write the geodesic equations for t and x :

$$\begin{aligned} \mu = t : \quad & \ddot{t} + \Gamma_{tt}^t \dot{t}^2 + 2\Gamma_{tx}^t \dot{t} \dot{x} + \Gamma_{xx}^t \dot{x}^2 = 0 \\ & \ddot{t} - \frac{1}{t} \dot{t}^2 - \frac{1}{t} \dot{x}^2 = 0 \\ \mu = x : \quad & \ddot{x} + \Gamma_{tx}^x \dot{t} \dot{x} + \Gamma_{xt}^x \dot{t} \dot{x} = 0 \\ & \ddot{x} - \frac{2}{t} \dot{t} \dot{x} = 0. \end{aligned}$$

Therefore, the geodesic equations for this spacetime are:

$$\begin{aligned}\ddot{t} - \frac{1}{t}\dot{t}^2 - \frac{1}{t}\dot{x}^2 &= 0 \\ \ddot{x} - \frac{2}{t}\dot{t}\dot{x} &= 0.\end{aligned}$$

□

Problem 2. Consider the following line element,

$$ds^2 = -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right) dt^2 + \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1} dr^2 + (r^2 + b^2) (d\theta^2 + \sin^2(\theta) d\varphi^2),$$

where M and b are positive, real constants. Note that we are using geometrical units where $G = c = 1$.

- (i) Write down the metric and the inverse metric in the (t, r, θ, φ) coordinate system.
- (ii) Write down the non-zero Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$ for this metric. *Hint: although there are 40 Christoffel symbols that are distinct by taking account of the symmetry under $\beta \leftrightarrow \gamma$, only 9 are non-zero.*
- (iii) Write the geodesic equation in this metric for a massive particle, i.e. write equations for $\ddot{t}$, $\ddot{r}$, $\ddot{\theta}$, $\ddot{\varphi}$ where the double-dots are with respect to proper time τ .
- (iv) Calculate the effective potential for timelike orbits for arbitrary M and b .
- (v) Do bound orbital solutions exist for $M = 0$? Why or why not?
- (vi) (★) Build a numerical solver for integrating the geodesic equation given an initial position ε and ℓ . Here, ε corresponds to energy density as written in class. Can you find any interesting geometries and/or trajectories?

Solution to (i). The metric components in the (t, r, θ, φ) coordinate system are given by:

$$g_{tt} = -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right), \quad g_{rr} = \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1}, \quad g_{\theta\theta} = r^2 + b^2, \quad g_{\varphi\varphi} = (r^2 + b^2) \sin^2(\theta).$$

All off-diagonal components are zero. Thus, the metric tensor can be written as:

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 + b^2 & 0 \\ 0 & 0 & 0 & (r^2 + b^2) \sin^2(\theta) \end{pmatrix}.$$

The inverse metric components can be computed as follows:

$$g^{tt} = -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1}, \quad g^{rr} = \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right), \quad g^{\theta\theta} = \frac{1}{r^2 + b^2}, \quad g^{\varphi\varphi} = \frac{1}{(r^2 + b^2) \sin^2(\theta)}.$$

Thus, the inverse metric tensor can be written as:

$$g^{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1} & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right) & 0 & 0 \\ 0 & 0 & \frac{1}{r^2 + b^2} & 0 \\ 0 & 0 & 0 & \frac{1}{(r^2 + b^2) \sin^2(\theta)} \end{pmatrix}.$$

□

Solution to (ii). After computing all the Christoffel symbols using the metric components from part (i), we find that the non-zero Christoffel symbols are:

$$\begin{aligned}\Gamma_{rt}^t &= \frac{1}{2}g^{tt}\partial_r g_{tt} = \Gamma_{tr}^t \\ \Gamma_{tt}^r &= -\frac{1}{2}g^{rr}\partial_r g_{tt}, \quad \Gamma_{rr}^r = -\frac{1}{2}g^{rr}\partial_r g_{rr}, \quad \Gamma_{\theta\theta}^r = -\frac{1}{2}g^{rr}\partial_r g_{\theta\theta}, \quad \Gamma_{\varphi\varphi}^r = -\frac{1}{2}g^{rr}\partial_r g_{\varphi\varphi} \\ \Gamma_{r\theta}^\theta &= \frac{1}{2}g^{\theta\theta}\partial_r g_{\theta\theta} = \Gamma_{\theta r}^\theta, \quad \Gamma_{\varphi\varphi}^\theta = -\frac{1}{2}g^{\theta\theta}\partial_\theta g_{\varphi\varphi} \\ \Gamma_{r\varphi}^\varphi &= \frac{1}{2}g^{\varphi\varphi}\partial_r g_{\varphi\varphi} = \Gamma_{\varphi r}^\varphi, \quad \Gamma_{\theta\varphi}^\varphi = \frac{1}{2}g^{\varphi\varphi}\partial_\theta g_{\varphi\varphi} = \Gamma_{\varphi\theta}^\varphi.\end{aligned}\quad \square$$

Solution to (iii). The only geodesic equation for the t coordinate is:

$$\ddot{t} + g^{tt}\partial_r g_{tt}\dot{t}\dot{r} = 0.$$

The geodesic equation for the r coordinate is:

$$\ddot{r} - \frac{1}{2}g^{rr}\partial_r g_{tt}\dot{t}^2 - \frac{1}{2}g^{rr}\partial_r g_{rr}\dot{r}^2 - rg^{rr}\dot{\theta}^2 - rg^{rr}\sin^2(\theta)\dot{\varphi}^2 = 0.$$

The geodesic equation for the θ coordinate is:

$$\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \sin(\theta)\cos(\theta)\dot{\varphi}^2 = 0.$$

The geodesic equation for the φ coordinate is:

$$\ddot{\varphi} + \frac{2}{r}\dot{r}\dot{\varphi} + 2\cot(\theta)\dot{\theta}\dot{\varphi} = 0.$$

Thus, the geodesic equations for a massive particle in this metric are:

$$\begin{aligned}\ddot{t} + g^{tt}\partial_r g_{tt}\dot{t}\dot{r} &= 0 \\ \ddot{r} - \frac{1}{2}g^{rr}\partial_r g_{tt}\dot{t}^2 - \frac{1}{2}g^{rr}\partial_r g_{rr}\dot{r}^2 - rg^{rr}\dot{\theta}^2 - rg^{rr}\sin^2(\theta)\dot{\varphi}^2 &= 0 \\ \ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} - \sin(\theta)\cos(\theta)\dot{\varphi}^2 &= 0 \\ \ddot{\varphi} + \frac{2}{r}\dot{r}\dot{\varphi} + 2\cot(\theta)\dot{\theta}\dot{\varphi} &= 0.\end{aligned}\quad \square$$

Solution to (iv). For a massive particle in a static, spherically symmetric metric,

$$ds^2 = -f(r)dt^2 + h(r)dr^2 + r^2(d\theta^2 + \sin^2(\theta)d\varphi^2),$$

the conserved quantities are:

$$E = f(r)\dot{t}, \quad L = r^2\dot{\varphi},$$

and we consider motion in the equatorial plane $\theta = \pi/2$. The timelike condition $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -1$ gives:

$$-1 = -f(r)\dot{t}^2 + h(r)\dot{r}^2 + r^2\dot{\varphi}^2.$$

Substituting the conserved quantities, we find:

$$\dot{r}^2 = \frac{E^2}{f(r)h(r)} - \frac{1}{h(r)}\left(1 + \frac{L^2}{r^2}\right).$$

Define $V_{\text{eff}}(r)$ such that:

$$\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \frac{E^2}{2},$$

giving us:

$$V_{\text{eff}}(r) = \frac{1}{2} \left(1 + \frac{L^2}{r^2} \right) \frac{1}{h(r)}.$$

In the case where $h(r) = 1/f(r)$, e.g. for the metric:

$$f(r) = 1 - \frac{2M}{r} + \frac{b}{r^2},$$

the effective potential becomes:

$$V_{\text{eff}}(r) = f(r) \left(1 + \frac{L^2}{r^2} \right) = \left(1 - \frac{2M}{r} + \frac{b}{r^2} \right) \left(1 + \frac{L^2}{r^2} \right).$$

This potential determines the radial motion of a timelike particle with energy E and angular momentum L . $\square$

Solution to (v). To determine whether bound orbits exist for $M = 0$, we examine the effective potential for timelike particles:

$$V_{\text{eff}}(r) = \left(1 - \frac{2M}{r} + \frac{b}{r^2} \right) \left(1 + \frac{L^2}{r^2} \right).$$

Setting $M = 0$, the potential reduces to:

$$V_{\text{eff}}(r) = \left(1 + \frac{b}{r^2} \right) \left(1 + \frac{L^2}{r^2} \right) = 1 + \frac{L^2 + b}{r^2} + \frac{bL^2}{r^4} \quad (6)$$

For a bound orbit to exist, the effective potential must have a local minimum at some finite $r > 0$, so that the radial motion oscillates between two turning points. We check for extrema by differentiating Equation (6) with respect to r :

$$\frac{dV_{\text{eff}}}{dr} = -2 \frac{L^2 + b}{r^3} - 4 \frac{bL^2}{r^5} = -\frac{2(L^2 + b)r^2 + 4bL^2}{r^5} \quad (7)$$

Setting Equation (7) to zero to find critical points, we have:

$$2(L^2 + b)r^2 + 4bL^2 = 0 \Rightarrow r^2 = -\frac{2bL^2}{L^2 + b}.$$

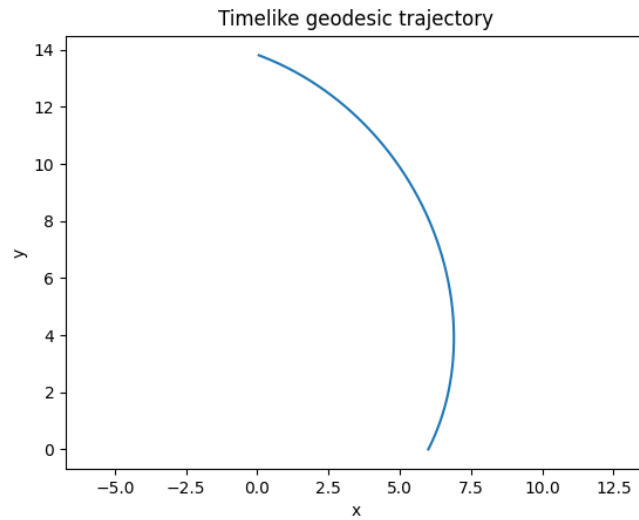
Since $r^2 > 0$, this equation has no real solutions for any real L and b , except possibly for negative b with $|b| < L^2$, which depends on the spacetime. In the usual case of positive b (repulsive term), there is no local minimum. Therefore, when $M = 0$, the effective potential is monotonically decreasing or always repulsive, and there is no finite radius at which a particle can be trapped. $\square$

Solution to (vi). To study the motion of timelike geodesics in the given spacetime, we numerically integrated the geodesic equations derived in part (iii). We worked in the equatorial plane $\theta = \pi/2$ without loss of generality, which reduces the system to the three variables $(t(\tau), r(\tau), \varphi(\tau))$ with initial energy density ε and angular momentum per unit mass ℓ .

The system of ODEs was evolved using a standard fourth-order Runge-Kutta integrator. From the numerical solution, we reconstructed the Cartesian coordinates

$$x(\tau) = r(\tau) \cos \varphi(\tau), \quad y(\tau) = r(\tau) \sin \varphi(\tau),$$

which give a spatial visualization of the orbit. A representative trajectory is shown below:



The resulting orbit displays a smooth inward-curving trajectory, typical of spacetimes with an effective attractive potential. The numerical integrator confirms the behavior predicted by the effective potential. For generic nonzero M a range of interesting timelike trajectories exists, while the $M = 0$ limit becomes purely repulsive in the radial direction and no bound orbits are observed, consistent with the analysis in part (v). $\square$